

Experiment: 7 Analog Beamforming Analysis Using MATLAB

Objective 1:

Generate beamforming gain values for different antenna array sizes using MATLAB. Compare how increasing the number of antenna elements improves the beamforming gain and enhances signal strength.

Objective 2:

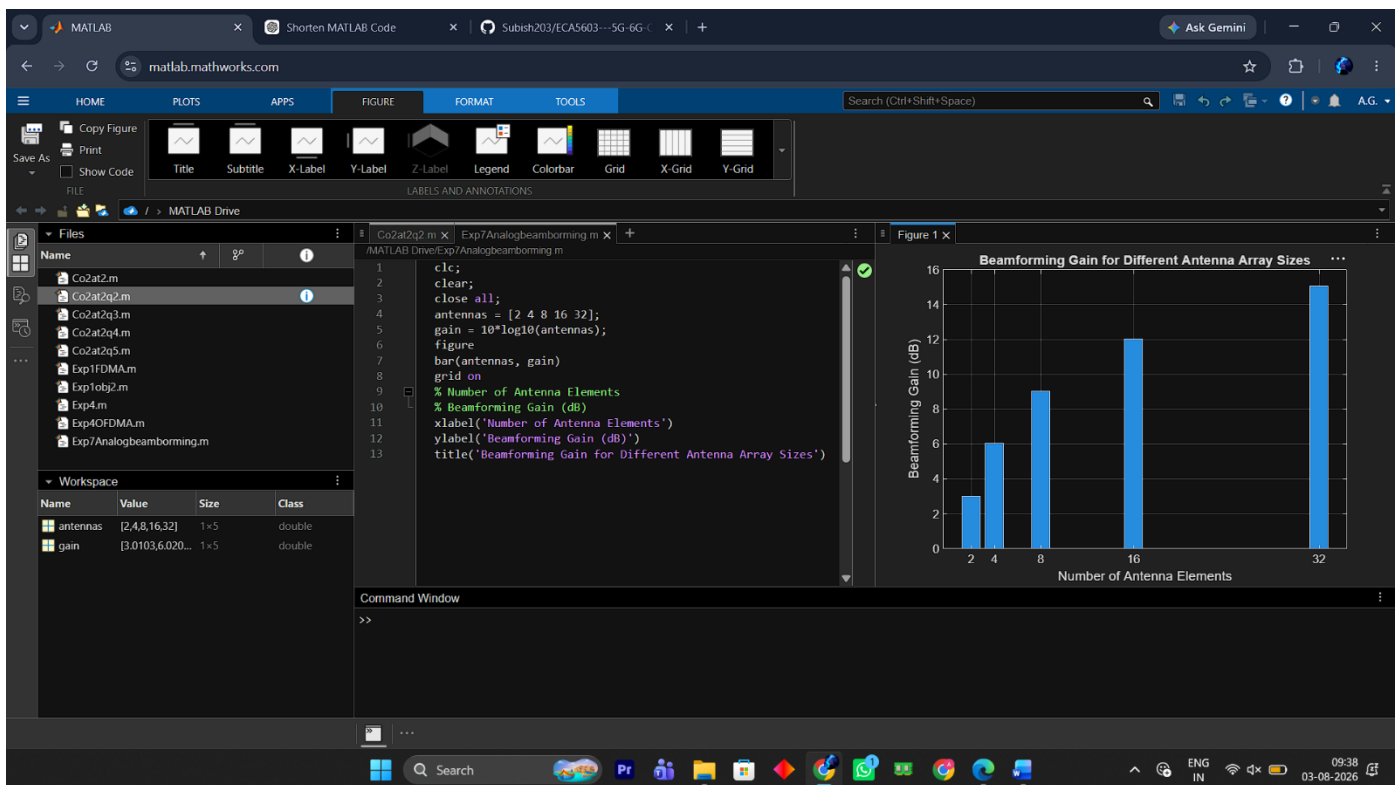
Analyze the beam width corresponding to different antenna array sizes using MATLAB. Observe how increasing the number of antenna elements results in a narrower beam, thereby improving spatial resolution.

Objective 3:

Simulate beam steering by varying the steering angle of the antenna array using MATLAB. Observe the change in the main lobe direction and understand how analog beamforming directs energy toward the intended user.

Objective 1 Matlab Code and Output:

```
clc;
clear;
close all;
antennas = [2 4 8 16 32];
gain = 10*log10(antennas);
figure
bar(antennas, gain)
grid on
% Number of Antenna Elements
% Beamforming Gain (dB)
xlabel('Number of Antenna Elements')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Antenna Array Sizes')
```



Objective 2 Matlab Code and Output:

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32];
```

```
% Number of Antenna Elements
```

```
beamwidth = [90 45 22.5 11.25 5.625];
```

```
figure
```

```
% Beam Width (degrees)
```

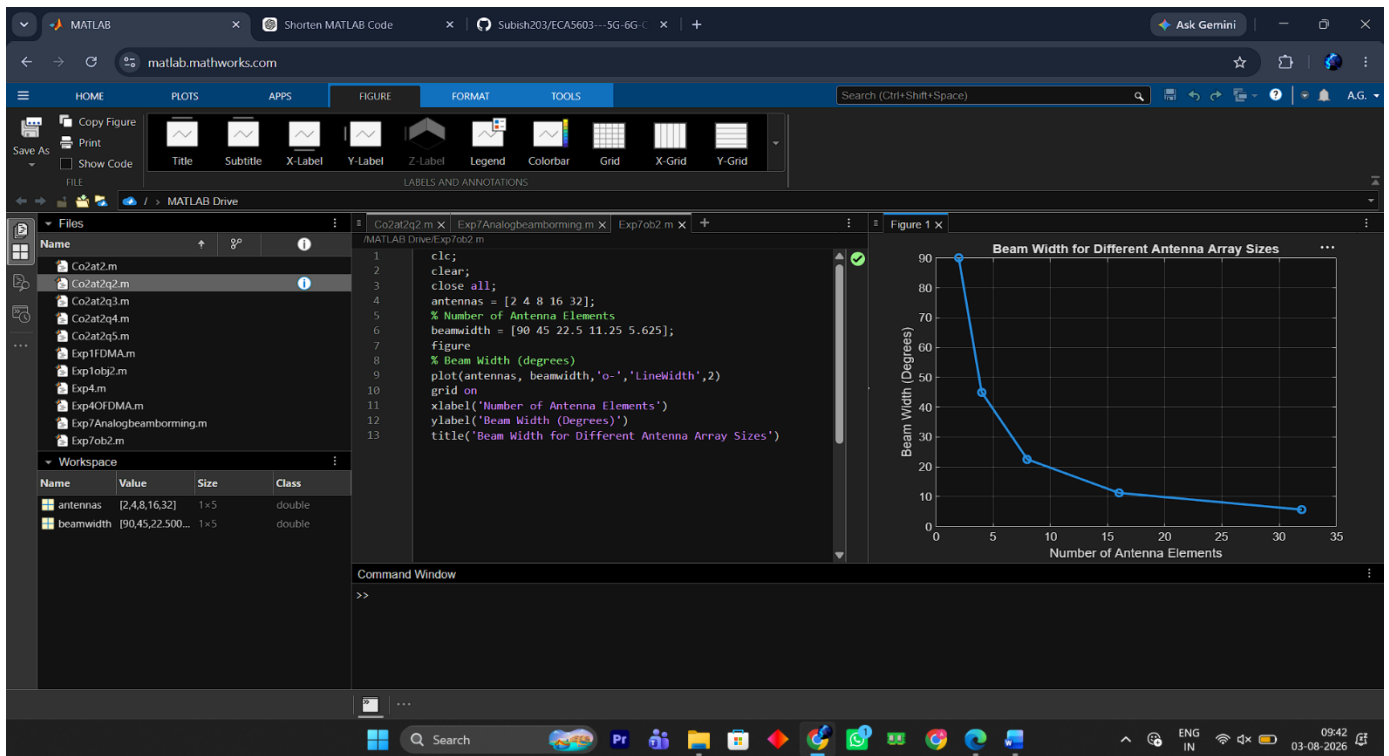
```
plot(antennas, beamwidth,'o-','LineWidth',2)
```

```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beam Width (Degrees)')
```

```
title('Beam Width for Different Antenna Array Sizes')
```



Objective 3 Matlab Code and Output:

```
clc;
```

```
clear;
```

```
close all;
```

```
theta = [-60 -30 0 30 60];
```

```
gain = [1 1 1 1 1];
```

```
figure
```

```
% Main Lobe Direction (degrees)
```

```
% Constant normalized gain
```

```
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
```

```
title('Main Lobe Direction in Analog Beamforming')
```

